

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech Branch : Electronics & Telecommunication Engineering and Allied

Semester: VI

Subject Code & Name: BTETPE603A Microprocessors and Microcontrollers

Max Marks: 60

Date: 23/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
	Which pin of the 8051 is used for providing the clock input?		1
	a. XTAL1 and XTAL2 b. RST c. PSEN d. ALE	CO1	
	The 8051 microcontroller has how many I/O ports?	CO1	1
	a. 4 b. 2 c. 8 d. 6		
3	How much on-chip RAM is available in the standard 8051 microcontroller?	CO1	1
	a. 64 bytes b. 256 bytes c. 512 bytes d. 128 bytes		
4	The instruction MOV A, @R1 uses which addressing mode?	CO1	1
	a. Immediate b. Direct c. Indirect d. Register		
5	Which flag is affected by the CMP instruction in the 8085?	CO1	1
	a. Auxiliary Carry Flag b. Zero Flag c. Sign Flag d. All of the above		
	What is the function of the HLT instruction in the 8085?	CO1	1
	a. Halt the program execution permanently b. Stop the microprocessor until reset c. Restart the program execution d. Enable the interrupt system		
7	Which of the following instructions is used to copy data from memory to the accumulator in 8085?	CO1	1
	a. LDA b. STA c. MOV d. MVI		
8	Which logical instruction complements all bits of the accumulator in 8085?	CO1	1
	a. CMA b. CMC c. XRA d. RLC		

9	What is the purpose of the CALL instruction?				CO1	1
	a. Halt the program execution	b. Jump unconditionally to a specified address	c. Call a subroutine and save the return address on the stack	d. Skip the next instruction		
10	Which instruction is used to push data onto the stack in 8085?				CO1	1
	a. POP	b. PUSH	c. CALL	d. HLT		
11	Which interrupt in 8085 has the highest priority?				CO1	1
	a. RST 7.5	b. RST 6.5	c. RST 5.5	d. TRAP		
12	In RISC processors, most instructions are				CO1	1
	a. Implemented using microcode	b. Executed in multiple cycles	c. Designed for hardware implementation	d. Supported with variable lengths		
Q. 2 Solve the following.						12
A)	Explain functions of registers in 8085 microprocessor.				CO2	6
B)	Explain the functions of each pin of 8085 microprocessors.				CO3	6
Solve the following.						12
A)	Write an 8085-assembly program to perform addition of two 8-bit numbers and store 16-bit result after addition in a specific memory locations.				CO2	6
B)	Write an 8085-assembly program to perform logical AND operation between two data bytes and store the result.				CO2	6
Q. 4 Solve Any Two of the following.						12
A)	Describe the internal memory organization of the 8051 microcontrollers.				CO1	6
B)	What are the key features of the 8051 microcontroller and explain the architecture of the 8051 microcontrollers.				CO1	6
	Define Addressing Mode? and explain addressing modes of 8051 microcontroller with examples.				CO3	6
Solve Any Two of the following.						12
A)	Write a program to generate a square wave of 50% duty cycle on the pin P1.5 of 8051. Use Timer 0 to generate the time delay.				CO2	6
B)	Describe functions of all bits of TCON register.				CO1	6
C)	Explain the basic concept and working of timers in the 8051 microcontrollers.				CO1	6

Q. 6	Solve Any Two of the following.		12
A)	Describe the functional blocks of the 8255 Programmable Peripheral Interface and explain how to interface it with the 8085 microprocessors.	CO3	6
B)	Draw the interfacing diagram of a 4K EPROM having a starting address 2000H with 8085 microprocessor using 3 to 8 Decoder (74LS138).	CO2	6
C)	Explain the working of an ADC and describe how it is interfaced with the 8051 microcontroller.	CO3	6
*** End ***			